Building Data Apps In Python With Streamlit

Session 3: Graphics

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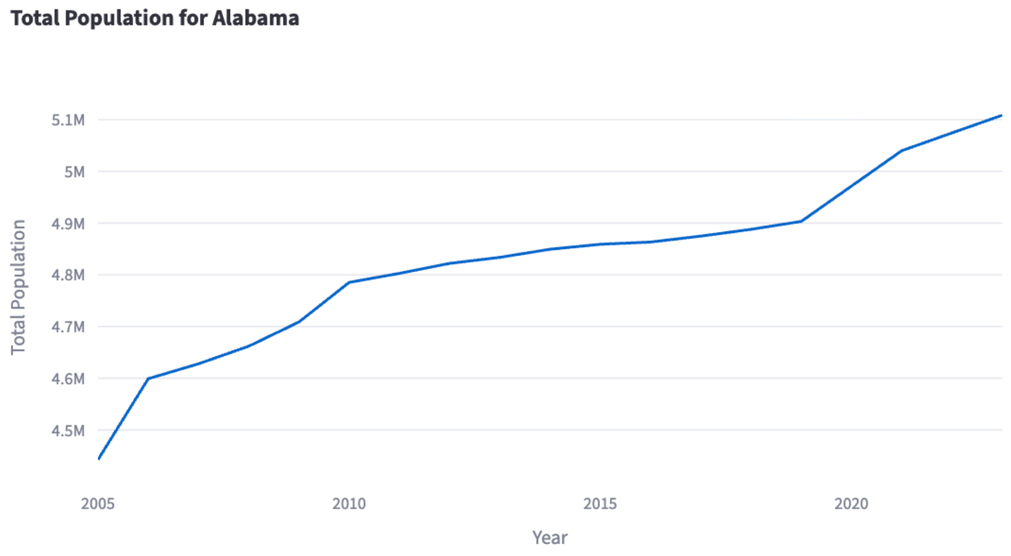
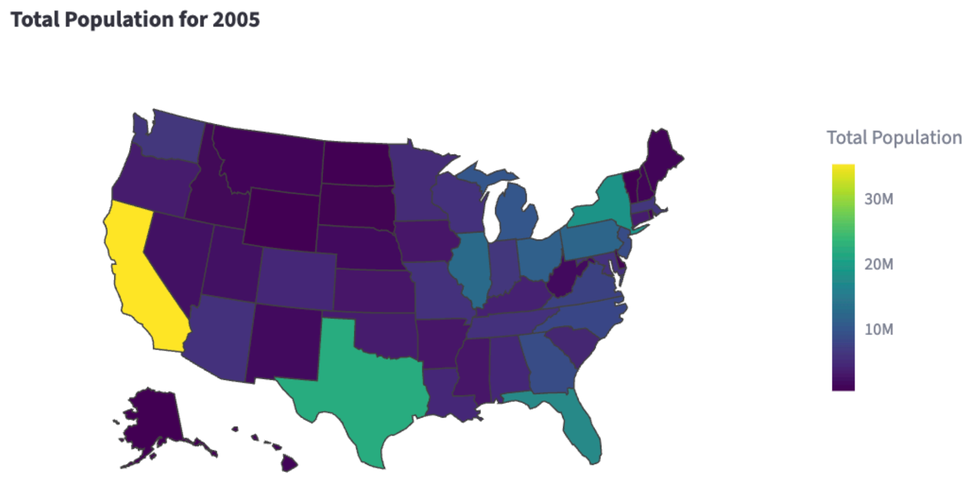
Course Agenda
(Big Picture)

1. Intro
2. Working with User Input
3. Creating Interactive Graphics





Exercise: What Graphics?

At the start of the first session I asked:

What do you want to build after completing this course?

Now answer this question:

What graphs do you want your app to have?

Please Open

- 3-graphics/graphs.ipynb
- 3-graphics/graph_app.py

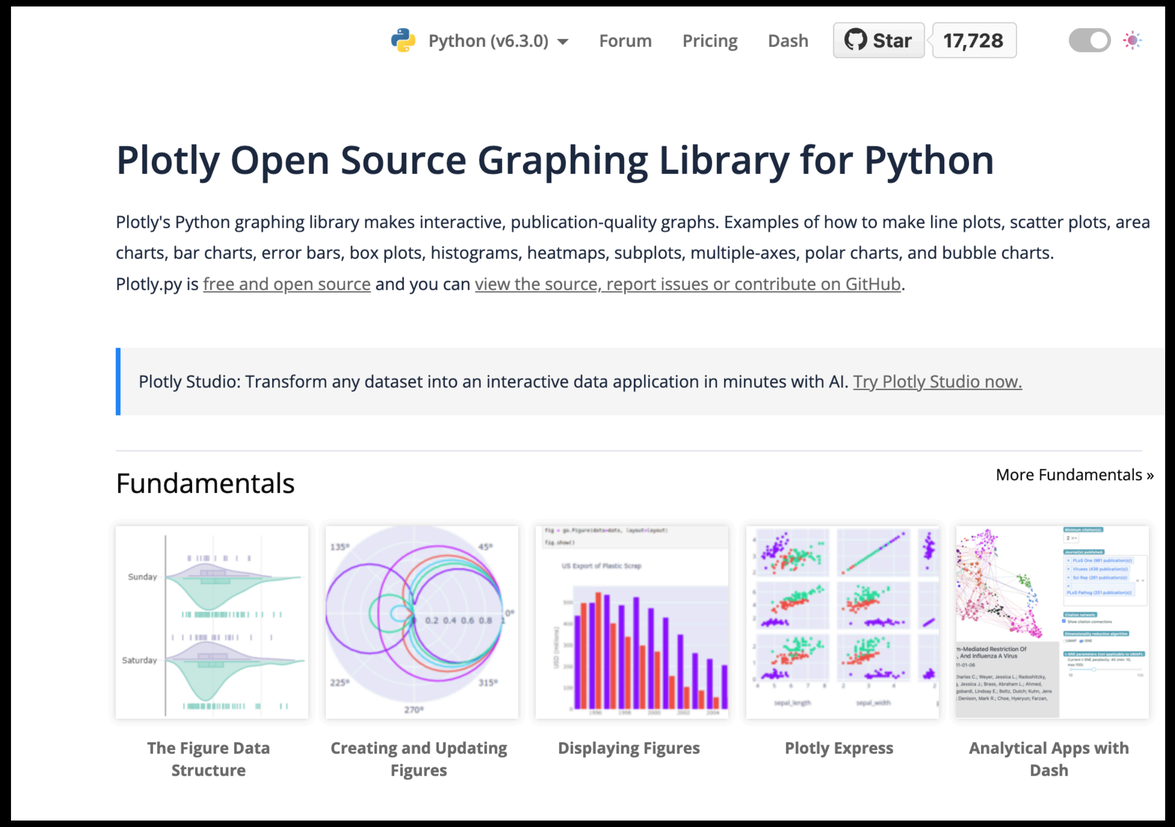
In the terminal, please also change your directory to 3-graphics

If you are not using VS Code to view the notebook, type:

```
uv run jupyter notebook graphs.ipynb
```

Plotly

- Popular library for interactive graphics
- "Plotly Express"
  - import plotly.express as px
  - px.line
  - px.choropleth

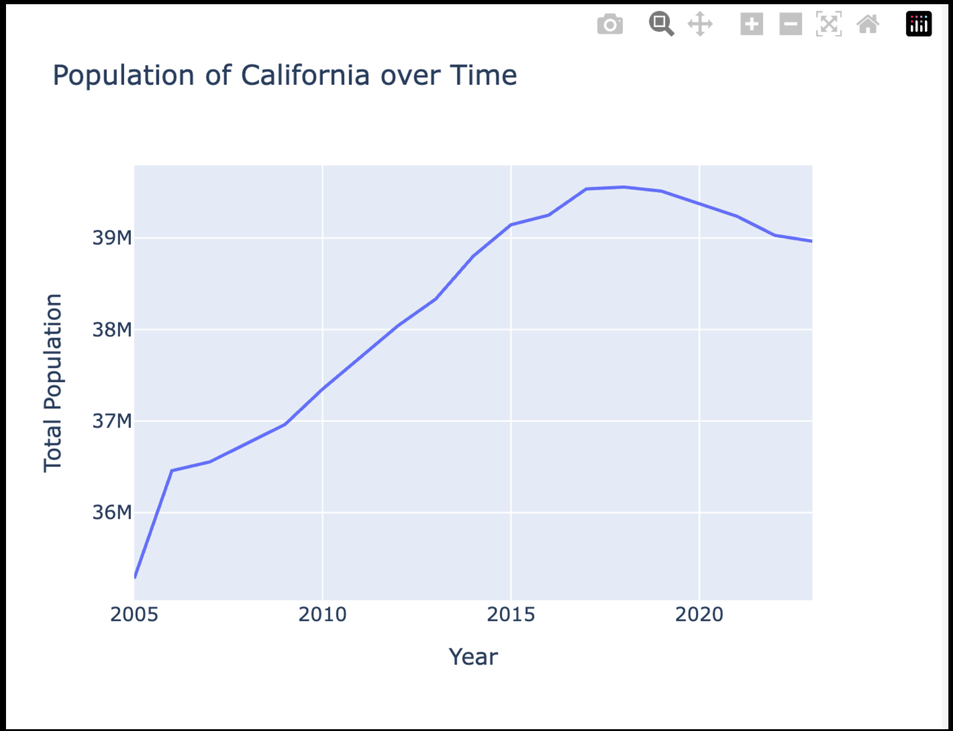


px.line()

Graph Population over Time

	State	Year	Total Population	Median Household Income
72	California	2005	35278768	53629
73	California	2006	36457549	56645
74	California	2007	36553215	59948
75	California	2008	36756666	61021
76	California	2009	36961664	58931
77	California	2010	37349363	57708
78	California	2011	37691912	57287

```python
import plotly.express as px

px.line(
    df_ca,
    x="Year",
    y="Total Population",
    title="Population of California over Time"
)
```



Plotly Graphs in Streamlit

st.plotly_chart

Version 1.49.0

Display an interactive Plotly chart.

Plotly is a charting library for Python. The arguments to this function closely follow the ones for Plotly's `plot()` function.

To show Plotly charts in Streamlit, call `st.plotly_chart` wherever you would call Plotly's `py.plot` or `py.iplot`.

Plotly Graphs in Streamlit

```
fig = px.line(...)
st.plotly_chart(fig)
```

f-Strings

f-Strings

For the next exercise, it will help to know about "f-strings".

```
state = "New York"
title = f"Graph for {state}"
title
```

✓ 0.0s

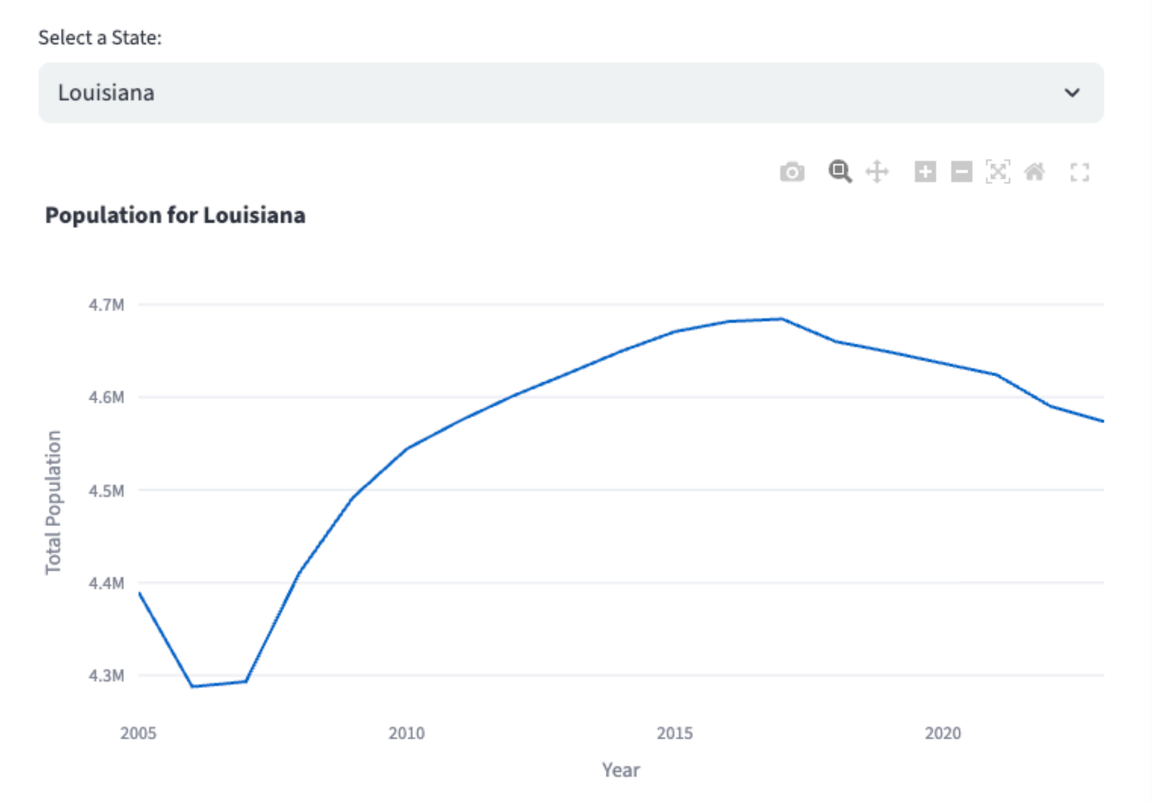
```
'Graph for New York'
```

Exercise: Plotly Graphs in Streamlit

Update `graph_app.py` to create a line graph of the population of the selected State. Use an f-string for the title of the graph. It should say the name of the state.

Solution

```
fig = px.line(df_state,
    x="Year",
    y="Total Population",
    title=f"Population for {state}")
st.plotly_chart(fig)
```



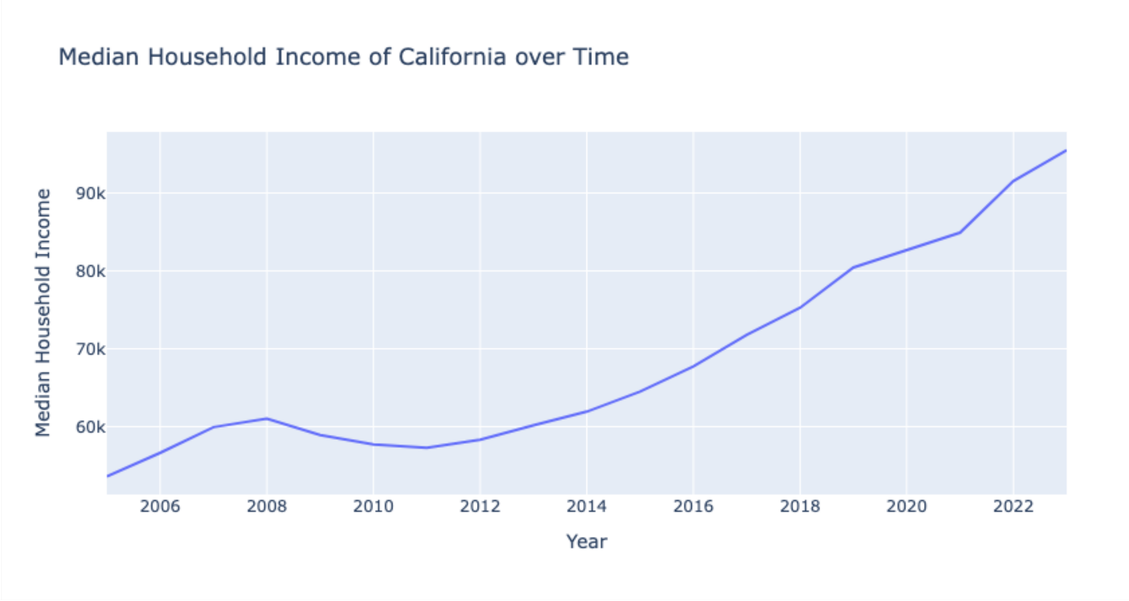
Two Columns: Population and Income

	State	Year	Total Population	Median Household Income	State Abbrev
0	Alabama	2005	4442558	36879	AL
1	Alabama	2006	4599030	38783	AL
2	Alabama	2007	4627851	40554	AL
3	Alabama	2008	4661900	42666	AL
4	Alabama	2009	4708708	40489	AL

Graphing Income

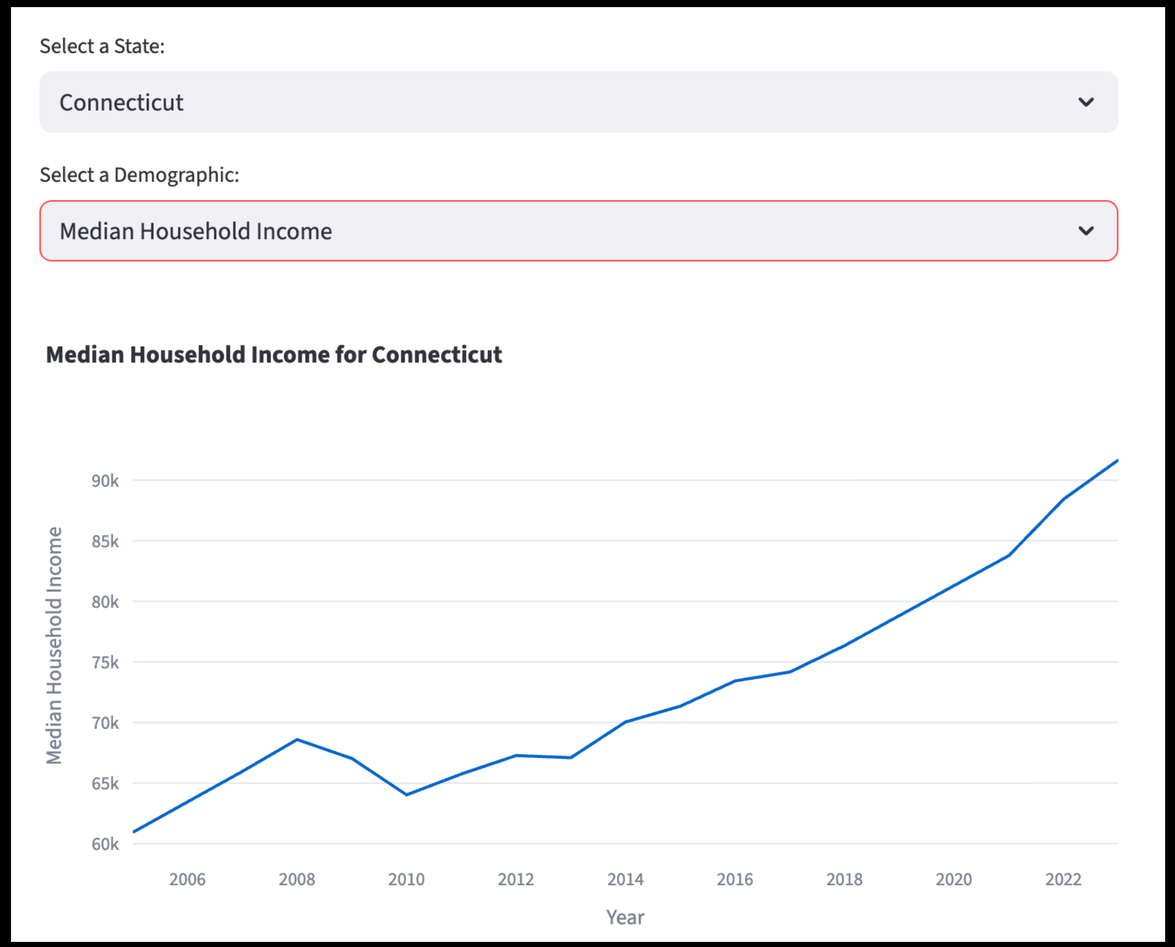
	State	Year	Total Population	Median Household Income	State Abbrev
0	Alabama	2005	4442558	36879	AL
1	Alabama	2006	4599030	38783	AL
2	Alabama	2007	4627851	40554	AL
3	Alabama	2008	4661900	42666	AL
4	Alabama	2009	4708708	40489	AL

```
px.line(
    df_ca,
    x="Year",
    y="Median Household Income",
    title="Median Household Income of California over Time",
)
```



Exercise: Select What to Graph

- Add a new select box to graph_app.py.
  - Options: ['Total Population', 'Median Household Income']
- Store the value returned from that selectbox in a variable called demographic.
- Use the value of demographic as the value for y in your line graph.

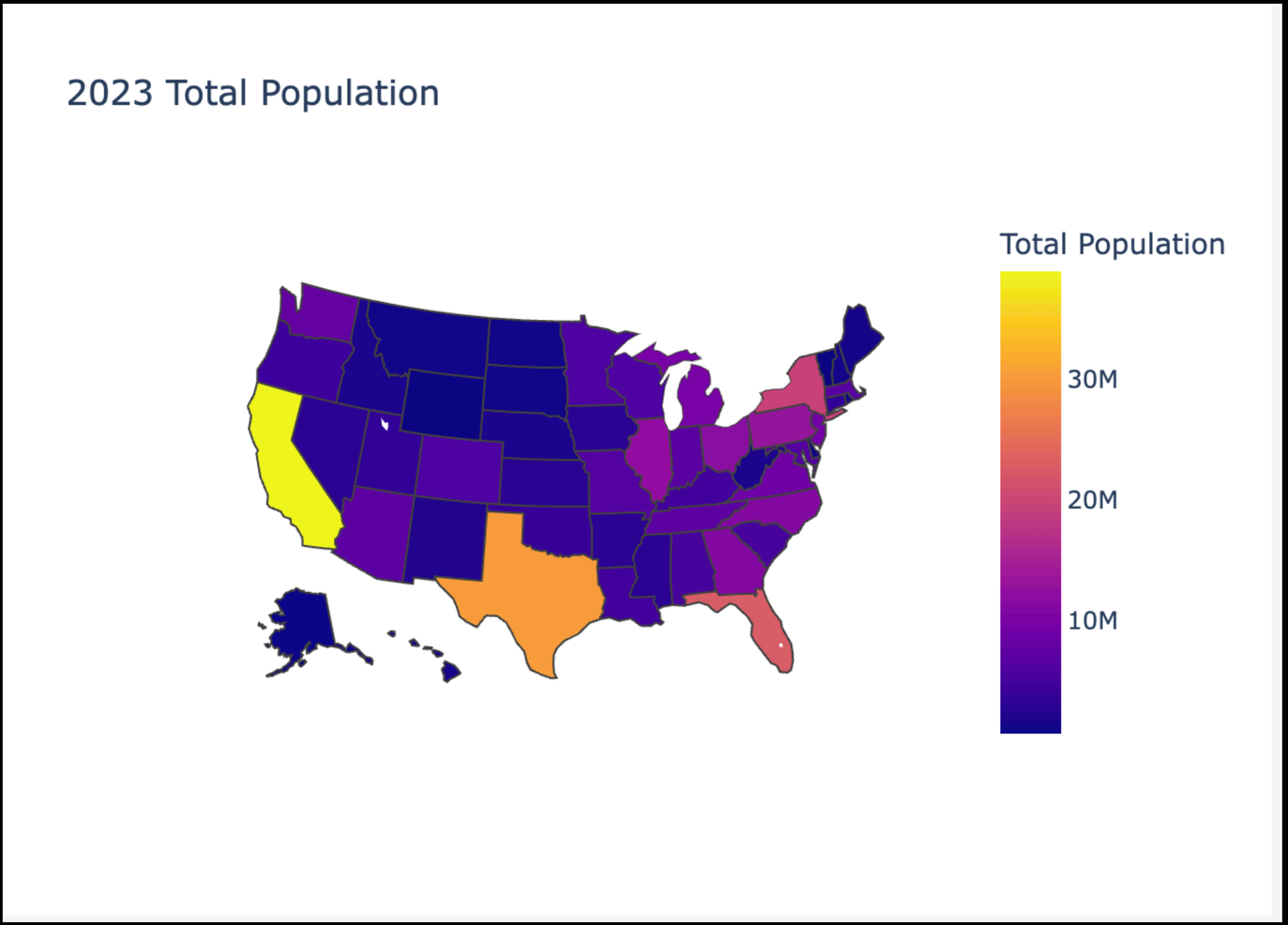


Solution

```python
import streamlit as st
import pandas as pd
import plotly.express as px

st.title("Demo Streamlit App")

df = pd.read_csv("state_data.csv")

state = st.selectbox("Select a State:", df["State"].unique())
demographic = st.selectbox(
    "Select a Demographic:", ["Total Population", "Median Household Income"]
)

mask = df["State"] == state
df_state = df[mask]

fig = px.line(df_state, x="Year", y=demographic, title=f"{demographic} for {state}")
st.plotly_chart(fig)

st.dataframe(df_state)
```

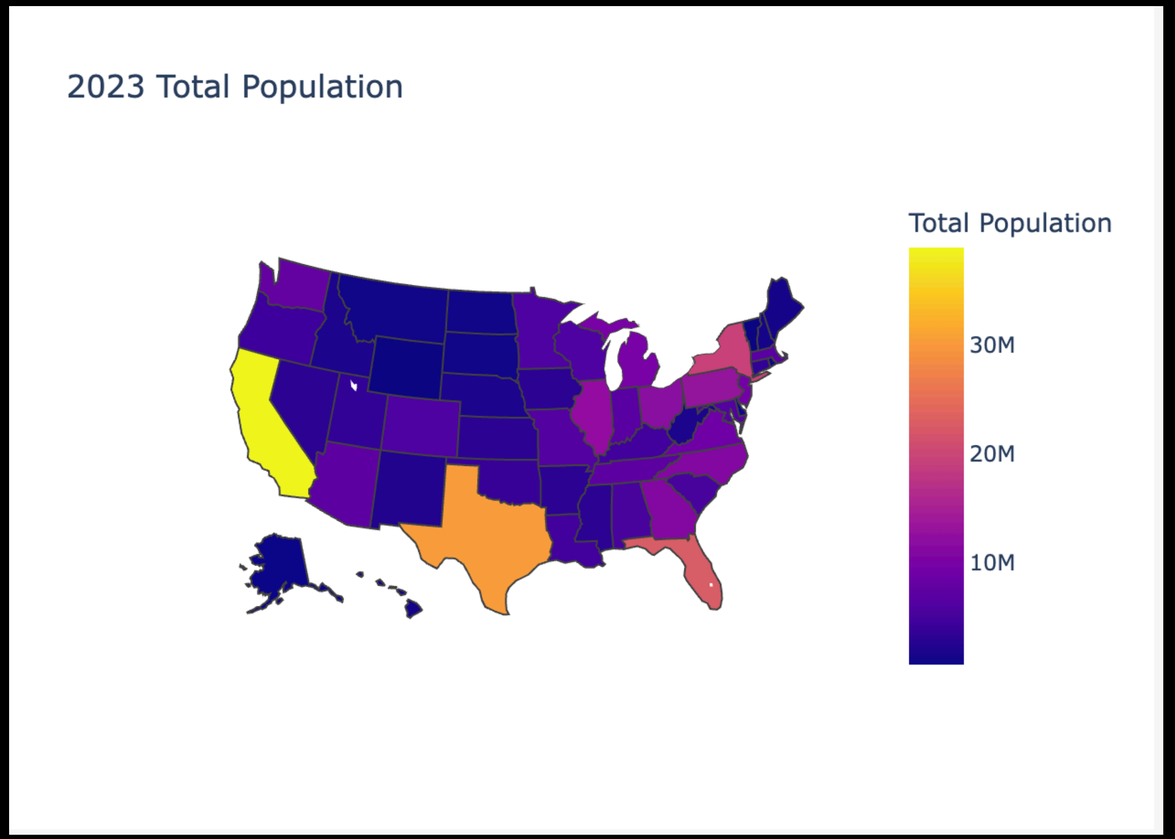
Choropleth Maps

- Shows locations such as states
- Uses color to express a value for each region.
- Different than a heat map



Example Choropleth Code

```python
mask = df["Year"] == 2023
df_2023 = df[mask]

px.choropleth(
    df_2023,
    locations="State Abbrev",  # Column for region
    locationmode="USA-states",
    color="Total Population",  # Column for color
    scope="usa",
    title="2023 Total Population",
    color_continuous_scale="viridis",
)
```

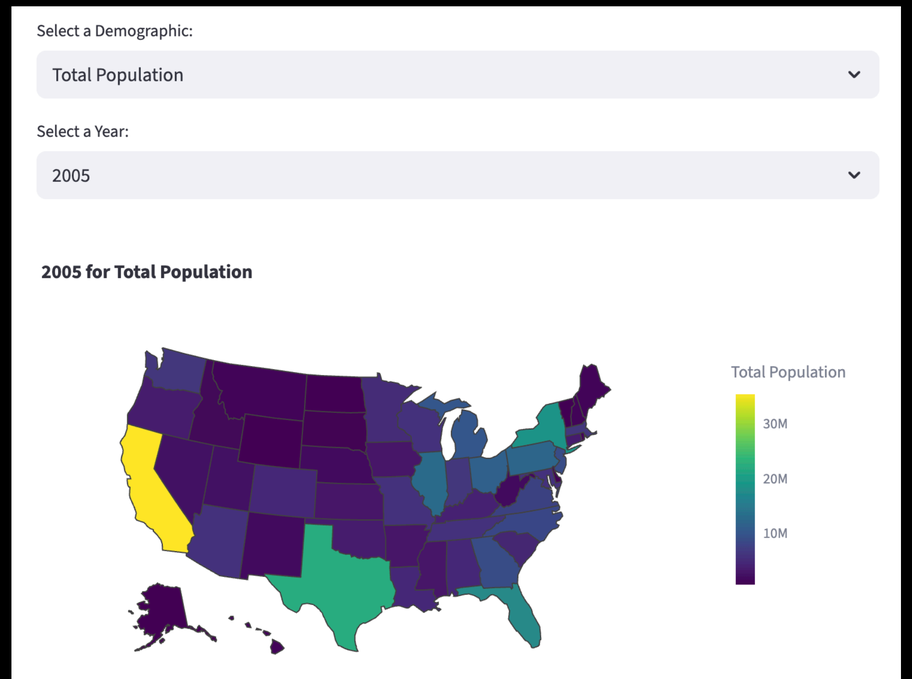


Exercise: Choropleth Map

1. Copy the above code, verbatim, to the app. Verify that it works.
2. Connect the map to the `demographic` selectbox. Do the results surprise you?
3. Create a new selectbox that lets the user select which `Year` of data to map.
4. Connect the year selectbox to the map.

The result should be an app that lets users select which year and demographic statistic to map.

Question: How does the Total Population map change over time? How about the Median Household Income map?



Solution

```
demographic = st.selectbox(
    "Select a Demographic:", ["Total Population", "Median Household Income"]
)
year = st.selectbox("Select a Year:", df["Year"].unique())

mask = df["Year"] == year
df_choro = df[mask]

fig = px.choropleth(
    df_choro,
    locations="State Abbrev",  # Column for region
    locationmode="USA-states",
    color=demographic,  # Column for color
    scope="usa",
    title=f"{year} {demographic}",
)
st.plotly_chart(fig)
```

Exercise: Copilot and Graphics

- Visit: copilot.microsoft.com
- Prompt: "What graphics would be good to add to this app? Provide example code."
  - Copy-paste the entire app into the prompt and hit enter. This should all be 1 prompt
- Evaluate the response.
- Add one of the suggestions to the app and click "Rerun".
  - Does it work? Is it helpful?

What graphics would be good to add to this app? Provide example code.

```
import streamlit as st
import pandas as pd
import plotly.express as px

st.title("Demo Streamlit App")

df = pd.read_csv("state_data.csv")

state = st.selectbox("Select a State:", df["State"].unique())
demographic = st.selectbox("Select a Demographic:", ['Total Population',
'Median Household Income'])
year = st.selectbox("Select a Year:", df['Year'].unique())

mask = df["State"] == state
df_state = df[mask]

fig = px.line(
    df_state,
```

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(Big Picture)

1. Session 1: Intro
2. Session 2: Working with User Input
3. Session 3: Graphics
4. Session 4: UI